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PHASE 1: PROJECT PROPOSAL

GROUP 8

Service-Oriented Smart Campus Support System

(Service-Oriented Architecture)

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1.0 Problem Description

Modern university campuses face challenges in managing student services due to fragmented and poorly integrated systems. Students and staff often use multiple platforms for tasks like course registration, facility booking, issue reporting, and receiving updates, which leads to inconvenience, duplicated data, and delays in accessing services.

For instance, course registration may lack real-time validation for prerequisites or seat availability, while facility bookings can cause scheduling conflicts without a centralized system. Issue reporting is also inefficient, with limited visibility on request status. Additionally, the absence of a unified identity and access control system creates security risks and inconsistent authorization across services. Weak communication between systems further results in delayed or missed updates.

To address these issues, a Service-Oriented Smart Campus Support System is proposed. By using Service-Oriented Architecture (SOA), the system integrates key services such as Identity Management, Academic Management, Facility Booking, Helpdesk, and Notification into a modular and interoperable structure. This approach improves efficiency, ensures data consistency, enhances user experience, and supports future scalability.

2.0 System Objectives

The purpose of this system is to:

- Provide a platform that allows students and staff to access campus services in one system without having to use separate systems.
- Improve the efficiency of campus facility management through a more organized system. Duplication or errors will be reduced.
- Facilitate communication between users and the system through an automatic notification and alert system. Users will get the latest information in real-time.
- Develop a flexible and easily expandable system using service-oriented architecture (SOA) approach that allows the system to be improved or expanded easily without affecting the entire system.

3.0 Selected Architecture Style

This Smart Campus Support System will be designed using the RESTful for the Service-Oriented Architecture to break the systems into five autonomous and loosely coupled services such as identity, academic, booking, helpdesk, and notification.

3.1 Key Design Principles

The system follows the Service-Oriented Architecture principles in the following things:

- **Statelessness:** Every time the service gets a request it has all the information it needs to process that request. This will help the system work well with a lot of users and makes it easier to manage the servers
- **Loose Coupling:** The services can work on their own so the user can update the Facility Booking part without messing with the Identity or Academic parts because of the minimal dependency it has.
- **Standardized Contracts:** All the services communicate with each other in a way using something called REST API interfaces and a format called JSON. This helps all the different parts of the system work together smoothly.
- **Service Autonomy:** Each service controls its environment, which helps the whole campus system work better.

3.2 Service Classifications

To ensure the business alignments and technical efficiency, the services are categorized as follows:

- **Business Services:** These services represent the main and the core of the campus activity such as the academic management for course registration and facility booking for the room reservations.
- **Utility Services:** This service provides the shared technical tasks that will use the synchronous for the identity and access to the system that needs the immediate response right away.
- **Entity Services:** This service will act as the bridge to the database for handling simple tasks for specific objects like student profiles or room details.

4.0 Initial Functional Requirements

The Service-Oriented Smart Campus Support System is divided into five initial functions. Each function is designed to be autonomous while providing users the specific capabilities via RESTful API endpoints.

4.1 Identity & Access

This function acts as the important authority for user verification and role management in the system.

- FR-101: The system shall allow students and staff to log in using their unique Username and password.
- FR-102: The service shall provide an endpoint for other services to verify if a user has "Student" or "Admin" privileges.
- FR-103: The system shall provide an endpoint to validate if a user's session is still active before allowing access to other services.
- FR-104: The system shall retrieve user profile data (Name, Student ID, and Role) upon request from authorized services.

4.2 Academic & Course

This service manages the academic catalog and the enrollment process for students.

- FR-201: The system shall provide a list of available courses, including course codes, names, and credit hours.
- FR-202: The system shall allow students to register for specific courses during the open registration period.
- FR-203: The service shall verify if a student meets the requirements for a course before allowing registration.
- FR-204: The system shall automatically close registration for a course once the maximum number of students is reached.
- FR-205: The service shall verify the student's active status via the Identity function before finalizing any course enrollment.

4.3 Facility Booking

This service manages the digital reservation of physical campus resources like discussion rooms and computer labs.

- FR-301: The system shall display a list of available rooms (Discussion Room A, Lab 1) along with their availability status.
- FR-302: Students shall be able to create, view, and cancel reservations for specific time slots.
- FR-303: Conflict Prevention: The system shall validate that a room does not have another booking for the same time period.
- FR-304: The system shall communicate with the Identity Service to verify the student's eligibility before confirming a reservation.

4.4 Campus Support & Helpdesk

This function handles the technical and administrative requests and allows students to report issues digitally.

- FR-401: Users shall be able to submit an issues report with a title, description, and priority level (Low, Medium, High).
- FR-402: Students shall be able to track the progress of their issues report("Pending," "In Progress," "Resolved").
- FR-403: Staff shall be able to assign the issues report to specific technicians for resolution.

4.4 Notification

This utility function will handle the communication and user feedback.

- FR-501: The system shall trigger an automatic "Booking Successful" alert once a reservation is recorded in the facility booking functions.
- FR-502: The system shall notify the student whenever the status of their support ticket changes (from "Pending" to "Resolved").
- FR-503: The system shall provide a real-time notification to Administrators when a new facility is added or a critical issues report is submitted.
- FR-404: The system shall alert students when the registration deadline is approaching.

5.0 Team Member Roles

Team Members	Roles
NOR SAERAH BINTI ISMAIL	Project manager
NOR AIN NATASYA BINTI KAMARULZAMAN	Frontend developer
CHE KU NURULHUDA BINTI CHE KU SAIFUL MUZAHAR	Backend developer